Workshop 3 Solutions

1. $\lambda x.(((((x\,x)\,y)((y\,z)\,w))\,x)(\lambda x.((x\,y)\,z)))$ (we didn't ask for parentheses)

2. $\lambda x.(((a\,b)\,c)(\lambda y.((x\,y)\,z)))$

3. $$\dfrac{\dfrac{}{(\lambda y.y)(\lambda y.y) \longrightarrow \lambda y.y}\,\beta \qquad \dfrac{}{\lambda y.y \longrightarrow^* \lambda y.y}\,\text{refl}}{(\lambda y.y)(\lambda y.y) \longrightarrow^* (\lambda y.y)}\,\text{step}$$

$$\dfrac{\dfrac{}{(\lambda x.xx)(\lambda y.y) \longrightarrow (\lambda y.y)(\lambda y.y)}\,(\beta)}{(\lambda x.xx)(\lambda y.y) \longrightarrow^* (\lambda y.y)}\,\text{step}$$

4. $FV(\lambda x.\lambda v.(\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w))$
$= FV(\lambda v.(\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w)) \setminus \{x\}$
$= (FV((\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w)) \setminus \{v\}) \setminus \{x\}$
$= (FV((\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w)) \setminus \{x,v\}$
$= (FV((\lambda y.y\,x)(\lambda y.w\,y)) \cup FV(\lambda w.w)) \setminus \{x,v\}$
$= (FV(\lambda y.y\,x) \cup FV(\lambda y.w\,y) \cup (FV(w) \setminus \{w\}) \setminus \{x,v\}$
$= ((FV(y\,x) \setminus \{y\}) \cup (FV(w\,y) \setminus \{y\}) \cup (\{w\} \setminus \{w\}) \setminus \{x,v\}$
$= ((\{y,x\} \setminus \{y\}) \cup (\{w,y\} \setminus \{y\}) \cup \emptyset) \setminus \{x,v\}$
$= (\{x\} \cup \{w\}) \setminus \{x,v\}$
$= \{x,w\} \setminus \{x,v\}$
$= \{w\}$

$BV(\lambda x.\lambda v.(\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w))$
$= BV((\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w)) \cup \{x,v\}$
$= ((BV(y\,x) \cup \{y\}) \cup (BV(w\,y) \cup \{y\}) \cup (BV(w) \cup \{w\})) \cup \{x,v\}$
$= \{y,w\} \cup \{x,v\}$
$= \{x,y,v,w\}$

$\lambda x.\lambda v.(\lambda y.y\,x)(\lambda y.w\,y)(\lambda w.w)$

5 $[\lambda y.y/x]((\lambda x.xy)(\lambda y.x)) = ([\lambda y.y/x](\lambda x.xy))([\lambda y.y/x](\lambda y.x))$

$= (\lambda x.xy)(\lambda y.[\lambda y.y/x]x)$

$= (\lambda x.xy)(\lambda y.\lambda y.y)$

6 $[y/x](\lambda y.xy) = \lambda w.[y/x][w/y](xy)$

$= \lambda w.[y/x](([w/y]x)([w/y]y))$

$= \lambda w.[y/x](xw)$

$= \lambda w.([y/x]x)([y/x]w)$

$= \lambda w.yw$

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$$\dfrac{\dfrac{}{(\lambda z.z)(\lambda w.w) \to^{CBV} \lambda w.w}\ \text{beta}}{(\lambda x.x)((\lambda z.z)(\lambda w.w)) \to^{CBV} (\lambda x.x)(\lambda w.w)}\ \text{appRight}$$

$$\dfrac{}{(\lambda x.x)((\lambda z.z)(\lambda w.w)) \to^{CBN} (\lambda z.z)(\lambda w.w)}\ \text{beta}$$

8

$$\dfrac{\dfrac{}{\lambda x.\lambda y.y \Downarrow \lambda x.\lambda y.y}\ \text{fun} \qquad \dfrac{\dfrac{}{\lambda z.z \Downarrow \lambda z.z}\ \text{fun} \qquad \dfrac{}{\lambda w.w \Downarrow \lambda w.w}\ \text{fun} \qquad \dfrac{}{\textcircled{1}\ \lambda z.z \Downarrow \lambda z.z}\ \text{fun}}{(\lambda z.z)(\lambda w.w) \Downarrow \lambda w.w}\ \text{app} \qquad \dfrac{}{\textcircled{2}\ \lambda y.y \Downarrow \lambda y.y}\ \text{fun}}{(\lambda x.\lambda y.y)((\lambda z.z)(\lambda w.w)) \Downarrow \lambda y.y}\ \text{app}$$

① $[\lambda z.z/w]w = \lambda z.z$

② $[\lambda w.w/x](\lambda y.y) = \lambda y.y$

$$\dfrac{\dfrac{}{\lambda x.\lambda y.y \Downarrow \lambda x.\lambda y.y}\ \text{fun} \qquad \dfrac{}{\textcircled{3}\ \lambda y.y \Downarrow \lambda y.y}\ \text{fun}}{(\lambda x.\lambda y.y)((\lambda z.z)(\lambda w.w)) \Downarrow \lambda y.y}\ \text{app}$$

③ $[(\lambda z.z)(\lambda w.w)/x](\lambda y.y) = \lambda y.y$

9 $\omega = \lambda x.xx \quad \Omega = \omega\omega = (\lambda x.xx)(\lambda x.xx)$

$(\lambda x.\lambda y.y)\,\Omega = \boxed{(\lambda x.\lambda y.y)((\lambda x.xx)(\lambda x.xx))}$

$(\lambda x.\lambda y.y)((\lambda x.xx)(\lambda x.xx)) \xrightarrow{CBN} \lambda x.\lambda y.y$

$(\lambda x.\lambda y.y)((\lambda x.xx)(\lambda x.xx)) \xrightarrow{CBV} (\lambda x.\lambda y.y)((\lambda x.xx)(\lambda x.xx))$

$\rightarrow \cdots$